

Beam Optics Modules

Optical design

- ▶ Accelerators differ by type, purpose, size, beam parameters:
 - ▶ Energy, luminosity, beam size, acceleration rate
 - ▶ This puts limits on the electromagnetic fields, dictates technology choices (e.g. normal vs superconducting, power consumption, etc.)
 - ▶ Aperture, momentum acceptance
 - ▶ Special requirements: source, diagnostics, collimators/halo removal, beam cooling, etc.
 - ▶ Number of users/experiments
 - ▶ Real estate: tunnel, beam lines, matching to existing infrastructure
 - ▶ etc., etc...

Optical design: examples

- ▶ Light source:
 - ▶ Synchrotron radiation due to bending
 - ▶ Low dispersion in the bending magnets
 - ▶ (small equilibrium emittances)
 - ▶ Relatively low energy
 - ▶ Many users
- ▶ HEP synchrotron:
 - ▶ High energy
 - ▶ Few experiments (but many users rely on them)
 - ▶ Lots of bending
 - ▶ Mostly FODO cells + space for RF and injection/extraction
 - ▶ Much large packing fraction (80%+ in the arcs)

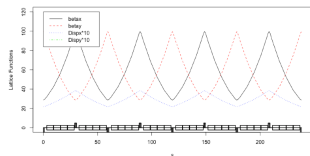
Optical modules

- ▶ Optical system building blocks
- ▶ Each module has a purpose + special conditions to satisfy:
 - ▶ general beam transport
 - ▶ achromats
 - ▶ large dispersion for momentum selection
 - ▶ charge selection
 - ▶ final focus
 - ▶ beam cooling
 - ▶ drift space for diagnostic equipment
- ▶ Large/long systems are best generated with stable periodic lens system (with or without bending)
- ▶ Longer spaces for instrumentation, RF, switching magnets, experiments
- ▶ May need to match one focusing structure into a different focusing structure
- ▶ Simultaneously trying to control $(\alpha, \beta, D)_{x,y}$ as well as x, y, z and x', y' of the ideal trajectory along the beam line
- ▶ Various computer codes aid in design

Examples of basic modules

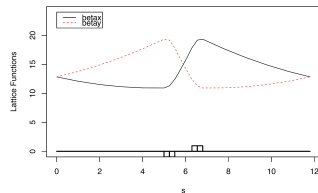
► FODO cells

- Basic transport
- Equal spacing
- Straightforward analysis
- Stability



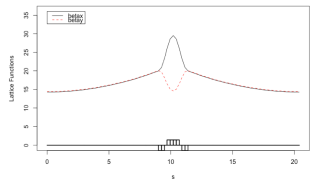
► Doublets:

- Used to generate more space
- As part of the more complex modules



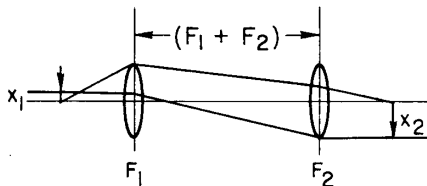
► Triplets:

- Can be used to keep the beam “round”
- As part of the more complex modules



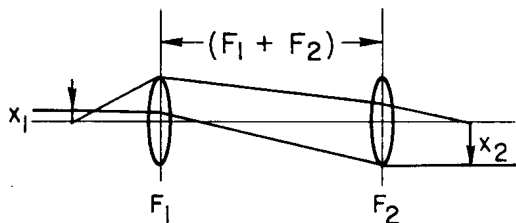
1D telescopic system

Used in colliders to achieve small beam size at IP (interaction point)



- ▶ Drift (f_1) – FQ (f_1) – Drift ($f_1 + f_2$) – FQ (f_2) – Drift (f_2)
- ▶ Notice:
 - ▶ both lens are focusing
 - ▶ we are using drift *lengths* that are matched to the focusing *strengths* of the lenses
- ▶ What does the system do as far as particle trajectories?
- ▶ What is the transfer matrix of the system?

1D telescopic system II

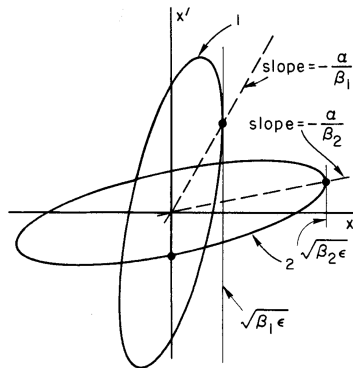


- ▶ $M = \begin{pmatrix} -m & 0 \\ 0 & -1/m \end{pmatrix}$, where $m = f_2/f_1$ - magnification

- ▶
$$\begin{pmatrix} \beta_2 \\ \alpha_2 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} m^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/m^2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \alpha_1 \\ \gamma_1 \end{pmatrix}$$

- ▶ $\beta_2 = m^2 \beta_1$, $\alpha_2 = \alpha_1 = \text{constant}$,
 $\tan \Delta\psi = \sin \Delta\psi = 0$

- ▶ Moreover, since $M_{11} = \sqrt{\beta_2/\beta_1} \cos \Delta\psi$, $\Rightarrow$
 $\cos \Delta\psi < 0$, $\Rightarrow \mu = \Delta\psi = \pi$



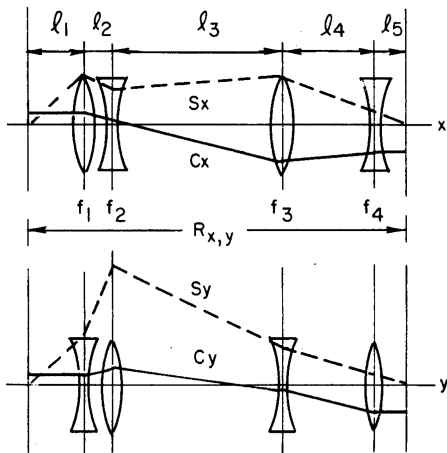
$\alpha = \text{constant}$

$$\beta_2 = M^2 \beta_1$$

$\Delta\psi = \pi = \text{Phase Shift}$

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2D telescopic system

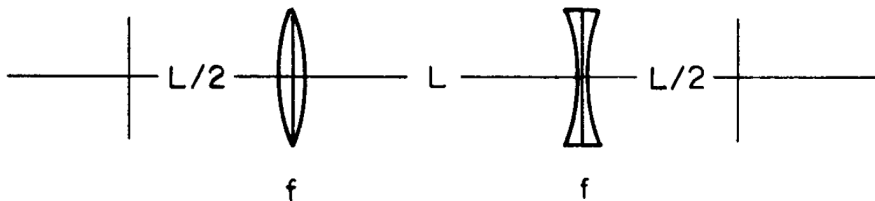


- ▶ To focus in both planes need at least two quadrupoles to simulate each lens (achieve focusing in both planes at each lens)

- ▶ $M = \begin{pmatrix} -m_x & 0 & 0 & 0 \\ 0 & -1/m_x & 0 & 0 \\ 0 & 0 & -m_y & 0 \\ 0 & 0 & 0 & -1/m_y \end{pmatrix}$

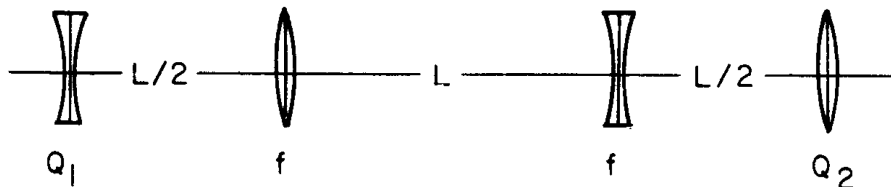
- ▶ m_x and m_y can be tuned independently (mind the trajectory deviation in the defocusing quads)

Quarter-wave transformer I



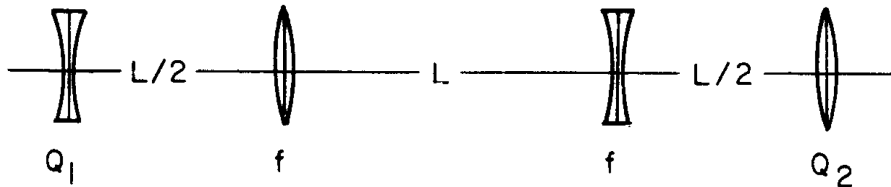
- ▶ Quarter-wave ($\lambda/4$) transformer is a special kind of FODO cell shown above
- ▶ **First, calculate the overall transfer matrix**
- ▶ From the transfer matrix one obtains: $\cos \mu = (1 - L^2/(2f^2))$,
 $\sin(\mu/2) = L/(2f)$
- ▶ So, $\beta_x = \beta_y = \frac{L}{\sin \mu} (2 - \sin^2(\mu/2))$ and $\alpha_x = -\alpha_y = -\frac{2 \sin(\mu/2)}{\sin \mu}$
- ▶ If we choose $\mu = \pi/2$ (quarter turn around the ellipse, hence, the name) then $L/f = \sqrt{2}$.
- ▶ **What is the transfer matrix for this particular case?**

Quarter-wave transformer II



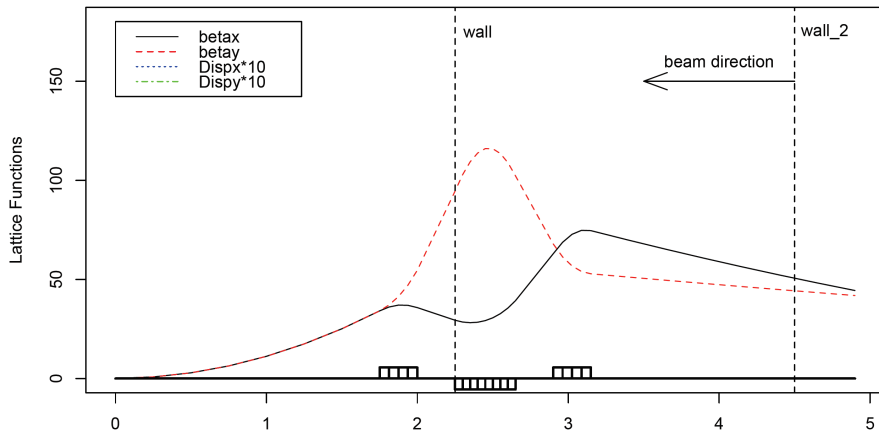
- ▶ From the previous calculation we obtain $M = \begin{pmatrix} \mp\sqrt{2} & 3L/2 \\ -2/L & \pm\sqrt{2} \end{pmatrix}$
- ▶ This is the right form for matching the two FODO cells with different parameters: $\beta_{1x} = \beta_{1y}$ and $\alpha_{1x} = -\alpha_{1y}$ to $\beta_{2x} = \beta_{2y}$ and $\alpha_{2x} = -\alpha_{2y}$
- ▶ The problem is that the quarter-wave cell will give you *some* (β_2, α_2) , not the ones you want
- ▶ Solution: add Q_1 and Q_2 on both sides of the cell and tweak those

Quarter-wave transformer III



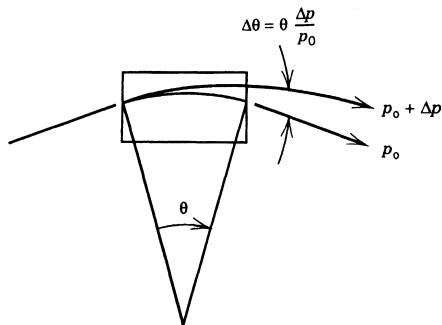
1. Quadrupole Q_1 does not change β_1 (so $\beta_{1x} = \beta_{1y}$ relation is safe)
2. Q_1 does not change the relation $\alpha_{1x} = -\alpha_{1y}$, but it does change the values of both
3. Adjust Q_1 such that when β_1 and α_1 input values are provided, the desired β_2 is achieved at exit (and some arbitrary value of α_2)
4. Keeping in mind that Q_2 does not change β_2 but does change α_2 , we can achieve the desired $\alpha_{2x} = -\alpha_{2y}$
5. Sometimes, Q_1 is not feasible, in this case one can tune L and f (remember, the only condition is $L/f = \sqrt{2}$ to get the correct β_2 and then proceed with tuning Q_2 as above.

Final focus



Starting at the target ($\beta = \beta^*$, $\alpha = \alpha^* = 0$), find values for tuning quads that match β_x, β_y back to the transport region (4 quads to match 4 parameters, $\beta_{x,y}, \alpha_{x,y}$)

Dispersion trajectory



[Slides on dispersion suppression:
courtesy of Onur Gilanliogullari.
See also: CERN'94 school intro material]

- ▶ The bend angle of a particle passing through a dipole magnet is

$$\theta = \frac{eB_0}{p} L,$$

B_0 – magnetic field, L – dipole length, $e/p = B\rho$ is the charge/momentum ratio or magnetic rigidity

- ▶ Particles with $p > p_0$ will be outside the design trajectory
- ▶ Particles with $p < p_0$ – inside
- ▶ The dependence of position on momentum is described by the dispersion function, $D(s)$:

$$\Delta x = D(s) \frac{\Delta p}{p_0}$$

Dispersion Relation

- ▶ We get dispersion relation from a modified version of Hill's equation that accounts for momentum variation:

$$x''(s) + \kappa(s)x = \frac{1}{\rho} \frac{\Delta p}{p}.$$

- ▶ The general solution for the inhomogeneous equation is a sum of a general solution for the homogeneous part and a particular solution for the inhomogeneous equation:

$$x(s) = x_0(s) + x_p(s).$$

- ▶ We can choose a particular solution as $x_p(s) = D(s) \frac{\Delta p}{p}$,
- ▶ where $D(s)$ satisfies

$$D''(s) + \kappa(s)D = \frac{1}{\rho}$$

- ▶ If we have a closed orbit, $\kappa(s + L) = \kappa(s)$, dispersion will also close onto itself.

Dispersion Relation

- ▶ General solution:

$$\begin{pmatrix} x \\ x' \end{pmatrix} = \begin{pmatrix} C & S \\ C' & S' \end{pmatrix} \begin{pmatrix} x_0 \\ x'_0 \end{pmatrix} + \frac{\Delta p}{p} \begin{pmatrix} D \\ D' \end{pmatrix},$$

- ▶ where

$$D(s) = S(s) \int_{s_0}^s \frac{1}{\rho} C(s) ds - C(s) \int_{s_0}^s \frac{1}{\rho} S(s) ds. \quad (1)$$

- ▶ Important observation: as $\rho \rightarrow \infty$, $D(s) \rightarrow 0$.
- ▶ Therefore, the only places dispersion can be created are those with local curvature (e.g. dipoles).

Dispersion Relation

- ▶ Looking at a dipole:

$$\mathcal{M} = \begin{pmatrix} \cos\left(\frac{l}{\rho}\right) & \rho \sin\left(\frac{l}{\rho}\right) \\ -\frac{1}{\rho} \sin\left(\frac{l}{\rho}\right) & \cos\left(\frac{l}{\rho}\right) \end{pmatrix}$$

- ▶ Therefore plugging it into Eq. (1):

$$D(s) = \rho \left(1 - \cos\left(\frac{l}{\rho}\right) \right),$$
$$D'(s) = \sin\left(\frac{l}{\rho}\right).$$

- ▶ After being created, the dispersion will get carried around focused and defocused by quadrupoles.

Dispersion Relation

- ▶ For the periodic dispersion function we know, $\eta(s + C) = \eta(s)$:

$$\begin{pmatrix} \eta \\ \eta' \\ 1 \end{pmatrix} = \begin{pmatrix} C & S & D \\ C' & S' & D' \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \eta \\ \eta' \\ 1 \end{pmatrix}.$$

- ▶ We can write this as

$$\begin{pmatrix} \eta \\ \eta' \end{pmatrix} = (I - \mathcal{M})^{-1} \cdot \begin{pmatrix} D \\ D' \end{pmatrix}$$

- ▶ Therefore,

$$\eta = \frac{(1 - S')D + SD'}{2(1 - \cos(\mu))},$$
$$\eta' = \frac{(1 - C)D' + C'D}{2(1 - \cos(\mu))}.$$

Dispersion Suppression

- ▶ How to suppress dispersion?
- ▶ The goal is to find a way to cancel dispersion without affecting Twiss parameters.
- ▶ We know dispersion is being focused in quadrupoles and created in dipoles.
- ▶ This implies we can find a way to insert extra dipoles and quadrupoles to cancel dispersion without affecting Twiss parameters.

Dispersion Suppression

- ▶ For a closed system, we can write a general matrix:

$$\mathcal{M} = \begin{pmatrix} \cos(\mu) + \alpha \sin(\mu) & \beta \sin(\mu) \\ -\gamma \sin(\mu) & \cos(\mu) - \alpha \sin(\mu) \end{pmatrix}.$$

- ▶ That gives us $\cos(\mu) = \frac{1}{2} \text{Tr}[M]$.
- ▶ Transformation matrix for a FODO cell:

$$\mathcal{M}_{\text{FODO}} = \begin{pmatrix} 1 - \frac{L^2}{8f^2} & L - \frac{L^2}{4f} \\ -\frac{L}{4f^2} \left(1 + \frac{L}{4f}\right) & 1 - \frac{L^2}{8f^2} \end{pmatrix}.$$

Dispersion Suppression

- ▶ By comparing the two representations, we obtain

$$\sin\left(\frac{\mu}{2}\right) = \frac{L}{4f}.$$

- ▶ We know how to calculate either the phase per cell or the focusing strength of the quadrupole.
- ▶ When it comes to dispersion, we need a 3×3 matrix.
- ▶ If the particular solution is η then

$$\begin{pmatrix} \eta \\ \eta' \\ 1 \end{pmatrix} = \begin{pmatrix} C & S & D \\ C' & S' & D' \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \eta_0 \\ \eta'_0 \\ 1 \end{pmatrix}.$$

Dispersion Suppression

- ▶ We need a 3×3 transformation matrix:

$$\mathcal{M}_{\text{FODO}} = \begin{pmatrix} 1 - \frac{L^2}{8f^2} & L(1 - \frac{L}{4f}) & \frac{L}{2}(1 - \frac{L}{8f})\theta_C \\ \frac{-L}{4f^2}(1 + \frac{L}{4f}) & 1 - \frac{L^2}{8f^2} & (1 + \frac{L}{8f}(1 - \frac{L}{4f}))\theta_C \\ 0 & 0 & 1 \end{pmatrix},$$

- ▶ where θ_C is total bending angle of the cell.
- ▶ We took the positions to be in the middle of both focusing and defocusing quadrupoles.
- ▶ The slope of dispersion should vanish at those points, therefore

$$\begin{pmatrix} \eta \\ 0 \\ 1 \end{pmatrix} = \mathcal{M} \cdot \begin{pmatrix} \eta_0 \\ 0 \\ 1 \end{pmatrix}.$$

Dispersion supression

- ▶ We get

$$\eta_{\pm} = \frac{L\theta_C}{4} \frac{1 \pm \frac{1}{2} \sin(\frac{\mu}{2})}{\sin^2(\frac{\mu}{2})}. \quad (2)$$

- ▶ Therefore, we can calculate the maximum and minimum of dispersion function.
- ▶ Now we need to cancel it:

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathcal{M}_{3 \times 3} \begin{pmatrix} \eta \\ 0 \\ 1 \end{pmatrix}. \quad (3)$$

Dispersion Suppression

- ▶ In order to keep Twiss parameters, we need to keep quadrupoles strength the same and add extra dipoles.
- ▶ If we introduce 2 FODO cells with different bending angles in each cell such that $\theta_C = \theta_1 + \theta_2$:

$$\mathcal{M}^{\theta_C} = M_{\text{FODO}}^{\theta_1} \cdot M_{\text{FODO}}^{\theta_2}. \quad (4)$$

- ▶ If we repeat the calculations, we can extract the following:

$$\theta_1 = \left(1 - \frac{1}{4 \sin^2 \frac{\mu}{2}} \right) \theta, \quad (5)$$

$$\theta_2 = \left(\frac{1}{4 \sin^2 \frac{\mu}{2}} \right) \theta. \quad (6)$$

Dispersion Suppression

- Or in a different way:

$$\theta_1 = \left(1 - \frac{1}{2(1 - \cos \mu)}\right) \theta, \quad (7)$$

$$\theta_2 = \frac{1}{2(1 - \cos \mu)} \theta. \quad (8)$$

- We can write it as

$$\theta_C = (1 - m)\theta + m\theta, \quad m = \frac{1}{2(1 - \cos \mu)} \quad (9)$$

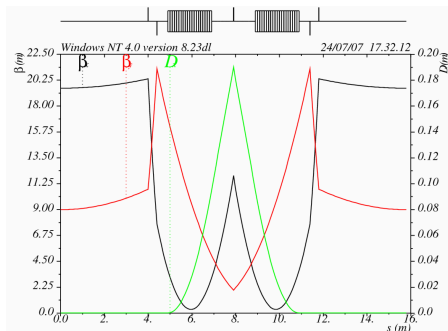
- If you choose $\mu = \frac{\pi}{2}$, then $\theta_1 = \theta_2 = \frac{1}{2}\theta$.

Dispersion Suppression

- ▶ Once dispersion is canceled, you can now add straight sections and the dispersion will remain zero.
- ▶ Let's look at some MAD-X code now...

Double bend achromat

- ▶ The Chasman–Green lattice aka DBA: a special periodic arrangement of magnets designed at BNL in the mid-1970s for synchrotrons.
- ▶ Provides optimized bending and focusing of electrons in storage rings designed for synchrotron light sources.
- ▶ Electron storage ring constructed with DBA: very low emittance, $\Rightarrow$ synchrotron light of exceptional brightness.
- ▶ Lattice of choice for most of the premier synchrotron light source facilities.
- ▶ Enter and exit with $D = D' = 0$
- ▶ Use two bend magnets with quad(s) in between
- ▶ If $d = 2f$ (distance between the quad lens and the bending magnets is twice the focusing distance) then $\Delta D'$ at lens: $\Delta D' = -d\theta/f = -2\theta$, where θ is the bending angle
- ▶ Since quad lens focuses in x , defocuses in y , we can use a triplet in the middle to control both planes



...and many other systems (chicane, bent solenoid, charge separator, chromaticity correction, etc.)